
Cp322 Machine learning project

By: Arbert Owusu, Jeryshan Varatheswaran

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Logical Partitioning/Split of Program Components

Group Member(s)	Component
Jeryshan	Preprocessing data
Arbert	Setting up training and testing sets and creating train and predicting functions
Arbert	Testing/evaluation/validation of regression models
Jeryshan	Testing/evaluation/validation of classification models
Arbert / Jeryshan	Visualization
Arbert	Feature engineering
Arbert / Jeryshan	Documentation, Editing, and other technical components

The Relevance of our Problem



Transportation delays are problems that commonly occur, so we decided to use the city of Toronto 2025 TTC (dataset) to predict delay in minutes and the delay risk.



We found this project to be useful for our understanding and our learning as we applied our understanding of ml concepts into a real-world issue. With this project we did not aim to solve transportation problems as that is unrealistic given the type of factors contributing to them. Rather, we aimed to better understand how metrics within the TTC, influence the likelihood of delays and the patterns associated with these delays.

Chosen Models / Methods

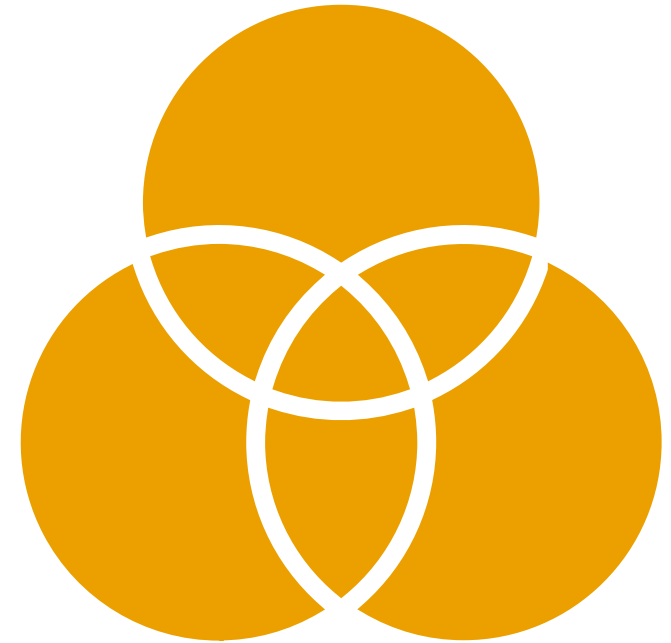
Regression Models	Classification Models
• Linear Regression model • Decision Tree regressor • Random Forest regressor	• Linear Discriminant Analysis (LDA) • Quadratic Discriminant Analysis (QDA) • Logistic Regression • Decision Tree (Classification) model • K-Nearest Neighbor (KNN) classification model



Why?

Linear Regression model

The linear regression model was chosen to figure out if the TTC dataset could be linearly separated. Even though we assumed otherwise because of sporadic events of transportation like weather, accidents, and infrastructure constraints. These were factors that we could not represent in the dataset. So, the linear model was used to primarily serve as a basis for how that model predicts the delay in minutes.



Decision Tree Regressor Model

The decision tree regressor model was chosen as we expected it to handle more complex data more appropriately. As linear regression model struggled to explain the data it was predicting. Suggesting to us that the data's underlying patterns was non-linear. So, we decided on the decision-tree since it uses a tree structure to hierarchically split the data to capture the best predictors. By doing this it could effectively capture non-linearity in the dataset to reduce variance and retain high speed given the size. Since, the data had high variability we expected it to perform better and based off metrics it had a significant improvement from the linear model.



Random Tree Regressor

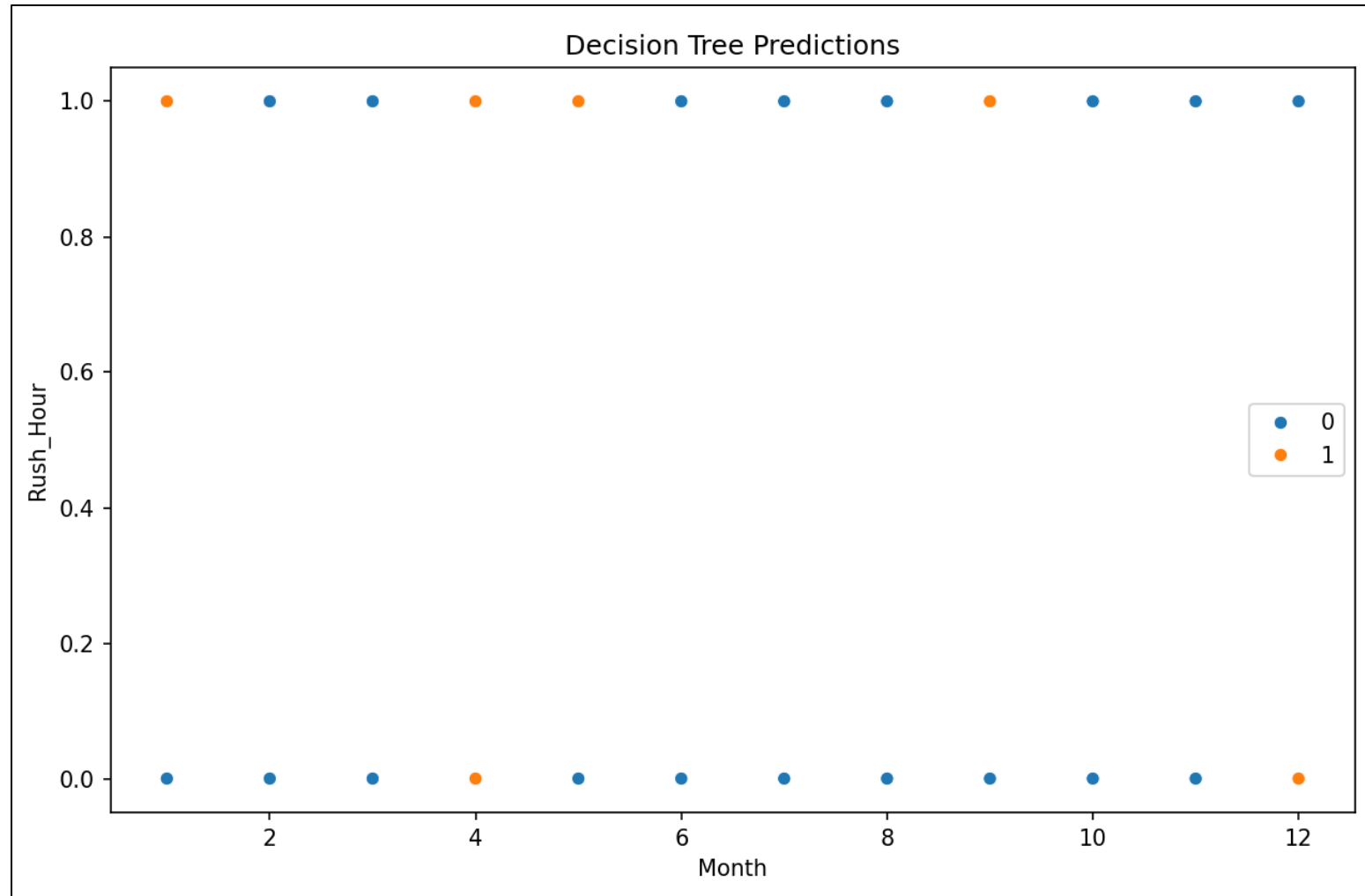
We chose the random forest regressor model to use multi tree-aggregation, to improve the predictability and reduce overfitting on a single tree as common in a single decision-tree model. However, a tradeoff we had to accept for the better stability and handling of noise was computationally. Since, the model uses many decision trees, it regressed the execution time but in return we did see improvements in our performance metrics.

Feature selection

For features we identified that useful features like the "Date" can be leveraged to engineer many features. From that feature we derived "Time_Group", "Day", "Hour", "Month", "Day_of_Week" and "Rush_Hour". With these extracted features we could remove the "Date" column as we had gained quantitative variables suitable for training. Alongside the already existing features after processing, these added features improved the model ability to capture patterns.

Decision Tree Classification Model & Findings

- Selected decision tree model over all other fitted classification models
- Decision Tree Accuracy Score: 80.56%
- 1 = High chance of delay risk (Equal to or more than 50%)
- 0 = Low or no chance of delay risk (Less than 50%)



The Correctness and Our Interpretations

Regression models	Mean Square Error	Root Mean Square Error	R^2
Linear Regression	83.66	9.15	0.03
Decision Tree Regression	61.16	7.82	0.29
Random Forest Regression	50.68	7.12	0.41

The Correctness and Our Interpretations

Classification Models	Accuracy
Linear Discriminant Analysis (LDA)	0.68
Quadratic Discriminant Analysis (QDA)	0.67
Logistic Regression	0.68
Decision Tree	0.78
K-Nearest Neighbour (KNN)	0.70

Grid Search for Decision Tree Model

Improved score: 0.8055561167199107

Improved parameters: {'max_depth': 20, 'min_samples_split': 64}

Best Decision Tree Model: DecisionTreeClassifier(max_depth=20, min_samples_split=64, random_state=42)

Best Decision Tree Parameters: {'max_depth': 20, 'min_samples_split': 64}

Completeness

Overall, we believe for predicting delay we could have explored more options such as XGboost tree to strengthen the complexity and improve predicting delay in minutes.

As XGBoost utilizes gradient descent to minimize the loss function iteratively across each new tree with new improvements to error. This would allow our model to capture the complexity of the dataset to extrapolate to predict more effectively.

Completeness

- Utilized several classification models for comparisons and analysis
- Useful techniques that could have been applied are regularization and PCA (Principal Component Analysis)
- Ensemble methods: Bagging, Bootstrapping, Boosting, etc.

Classification Models	Brief Description
Linear Discriminant Analysis (LDA)	A supervised machine learning algorithm that sets up linear decision boundaries.
Quadratic Discriminant Analysis (QDA)	A supervised machine learning algorithm that sets up non-linear decision boundaries.
Logistic Regression	A machine learning algorithm that predicts the probability of categorical outcomes based on input data.
Decision Tree (Classification Tree)	A machine learning method that categorizes data and converges upon certain points.
K-Nearest Neighbour (KNN)	An algorithm that checks a data point's neighbors to make a decision. Classifying a point of data based on the nearest k points or data, this eventually results in a boundary or several boundaries.

Findings of our Problem



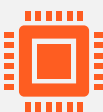
Non-linearity of the data



Tree-based top performers, specifically random-forest regressor for regression, decision tree for classification



Lower variability resulted in significant improvements to accuracy



The data benefited significantly from feature creation, and more features could serve as a valuable improvement to our models' performance

Creativity

We kept our project within the scope of the course, but we used a combination of techniques to achieve useful parameters for our project. Like integration of grid-search algorithm's to fine tune our models' parameters and kfold-cv to evaluate our classification models to find the average accuracy. Feature engineering to derive our own features and evaluating these features on multiple models. Also, we implemented this project in mind to be practical as this project involved finding patterns and delay-periods to improve predictability to classify and predict delay.

